MILITARY SPECIFICATION

DYE, B-1, WATER DISPERSED FORMATION

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water dispersed formation of B-1 dye (1-4'nitrophenylazo-2-naphthaleneamine).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations.

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 B-1 dye.

3.1.1 Chemical and physical characteristics. The B-1 dye, prior to being placed in the dispersed form, shall conform to the requirements of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Min	Max	
Free beta-naphthylamine, percent by weight	---	0.1	4.2.4.1
Solubility in acetone, grams per milliliter	0.04	0.05	4.2.4.2
Melting point, °C	178.0	187.0	4.2.4.3
Particle size:			
Percent retained on No. 325 sieve	---	0.5	4.2.4.4(a)
Percent less than 10 micrometers	10	20	4.2.4.4(b)
Mean mass diameter, in micrometers	15	30	4.2.4.4(b)
Purity of B-1 dye, percent of standard	98	102	4.2.4.7

3.1.1.1 Solubility in heptane. When tested as specified in 4.2.4.5, the plotting of absorbance versus concentration shall be a straight line and the absorbance range of a 3-milliliter (mL) aliquot shall be 0.07 to 0.13.

3.1.1.2 IR spectra, between 2 to 15 micrometers. All major bands shall be within $\pm$ 1 cm^{-1} of the specified bands when tested as specified in 4.2.4.6.

3.1.2 Standard sample. A 500-gram (g) dry sample of the B-1 dye from each lot (4.2.1), packaged in separate containers, shall be marked with the lot number and shipped to the address specified in 6.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the water dispersed formation of B-1 dye which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling. See 6.7 for sampling and testing precautions.

4.2.2.1 For examination of packaging. Each pack of water dispersed formation of B-1 dye shall be examined in accordance with the classification of defects.

4.2.2.2 For test. After washing and cleaning but prior to being placed in the dispersed form, a sufficient number of specimens shall be randomly taken from the in process B-1 dye to yield approximately 1000 g after drying. The specimens shall be consolidated and placed in a clean, dry container labeled to identify the lot from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. Each pack shall be examined for the following defects. Any defect shall be cause for rejection of the pack.

(a) Pack not as specified
(b) Marking not as specified
(c) Lot or batch number missing or incorrect
(d) Date of manufacture missing or incorrect

4.2.3.2 For test. See 6.7 for sampling and testing precautions. Particle size and mean mass diameter (4.2.4.4) shall be determined on the consolidated sample (4.2.2.2) as received. The remaining consolidated sample shall be dried and a representative 500 g forwarded as specified in 3.1.2. The remaining dried sample shall be tested in duplicate to determine conformance to the remaining requirements. If either duplicate fails any test, or if the particle size or mean mass diameter are not as required, the lot of B-1 dye represented shall not be used for the water dispersed formation.

4.2.4 Tests. See 6.7 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Free beta-naphthylamine.

(a) Preparation of calibration curve. Prepare a buffer solution of pH 4.0 by adding 2.0 mL of aqueous 0.1N sodium hydroxide to 250 mL of 0.1M potassium acid phthalate in a 500-mL volumetric flask. Mix well and make up to

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volume with water. Weigh 0.1 g of purified and recrystalized beta-naphthylamine to the nearest 0.1 milligram (mg) into a 500-mL volumetric flask. Add 100 mL of hydrochloric acid (aqueous solution, 1-percent by volume) and warm the mixture until bubbles start to rise. Stopper the flask and shake mechanically while the solution cools to room temperature. Make up to volume with the hydrochloric acid solution. Pipet a 50-mL aliquot into a second 500-mL volumetric flask and make up to volume with the hydrochloric acid solution (Solution A). From Solution A, prepare 5 solutions as follows: pipet 30-, 25-, 20-, and 10-mL aliquots into separate 50-mL volumetric flasks; pipet a second 10-mL aliquot into a 100-mL volumetric flask. Make each up to volume with the hydrochloric acid solution (Solutions C, D, E, F, and G, respectively). Treat each solution (C, D, E, F, and G) and a blank in duplicate as follows: pipet a 2-mL aliquot into a spectrophotometer cell, add 3.0 mL of buffer solution and 1.0 mL of absolute alcohol and mix (for the blank, use a 2-mL aliquot of the hydrochloric acid solution). Zero the spectrophotometer (equipped with a No. 52 filter) with water. Measure and record the absorbance of each of the duplicate solutions (for C, D, E, F, G, and blank). Add 1.0 mL of a 0.1 percent by volume naphthanil diazo blue B (NDBB) aqueous solution, prepared just before use, to each duplicate solution and mix. Measure and record the absorbance of each solution at exactly 5 minutes after the addition of the NDBB solution. Calculate the net absorbance of the blank as follows:

Net absorbance of blank = Average of 2d absorbance reading (after addition of NDBB) duplicates - 0.857 times average of 1st reading duplicates.

Calculate the net absorbance of each standard solution (Solutions C, D, E, F, and G) as follows:

$$A_S = Y - 0.857(X) - A_B$$

where: A_S = Net absorbance of standard solution,
X = Average of 1st absorbance readings of standard solution duplicates,
Y = Average of 2d absorbance readings of standard solution duplicates (after addition of NDBB), and
A_B = Net absorbance of blank.

Prepare a calibration curve of the net absorbances versus the known concentration (micrograms of beta-naphthylamine per 2.0 mL) for each of the net absorbances. Pass a straight line through the origin and the plotted points.

(b) Procedure. Weigh 0.1 g of B-1 dye specimen to the nearest 0.1 mg in a 100-mL volumetric flask. Add 50 mL of hydrochloric acid solution and warm the mixture until bubbles start to rise. Stopper the flask and shake mechanically while the solution cools to room temperature. Filter the solution through a sintered glass funnel, using a low vacuum. Rinse the flask with three 5-mL portions of hydrochloric acid solution and add to the funnel after

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each rinsing. Return the filtrate to the 100-mL volumetric flask. Rinse the filter flask with three 5-mL portions of hydrochloric acid and add to the volumetric flask after each rinsing. Make up to volume with hydrochloric acid solution and mix well. Pipet duplicate 2-mL aliquots into spectrophotometer cells and treat as above for the calibration curve procedure. Calculate net absorbance as above. From the calibration curve, determine beta-naphthylamine concentration. Calculate percent free beta-naphthylamine as follows:

$$\text{Percent free beta-naphthylamine} = \frac{5000C}{W}$$

where: C = Concentration of beta-naphthylamine, in micrograms per 2.0 mL, and
W = Weight, in grams, of B-1 dye specimen.

4.2.4.2 Solubility in acetone.

(a) Preparation of calibration curve. Weigh 0.009 to 0.010 g of standard B-1 dye (see 6.6) to the nearest 0.1 mg and dissolve in acetone in a 1-liter volumetric flask at 30°C (Solution A). From Solution A, prepare 5 additional solutions as follows: pipet 40-, 30-, 20-, and 10-mL aliquots of Solution A into separate 50-mL volumetric flasks and make each up to volume at 30°C with acetone (Solutions B, C, D, and E, respectively). Pipet a second 10-mL aliquot into a 100-mL volumetric flask and make up to volume at 30°C with acetone (Solution F). Determine percent transmittance of Solutions A, B, C, D, E, and F with a Beckman DU Visible Spectrophotometer or equal using regular visible cells of optical path 1 centimeter. Use acetone as a blank. Set slit width at S = 0.064 mm, = 510 mm. Measure the percent absorbance by first nulling the acetone blank at T = 100 percent and then measure each solution (A, B, C, D, E, and F) at least twice. Plot the absorbance along the log-axis versus concentration on 2 cycle semilog paper.

(b) Sampling apparatus. Mount a 125-mL Erlenmeyer flask in a constant temperature bath at 30°C. Mount the bath on a magnetic stirring apparatus capable of stirring the flask contents. A two-hole rubber stopper, one hole fitted with a right angle glass tube and the other hole fitted with a bulb filtering device, is placed on the flask (the right angle glass tube outside the flask; the bulb filtering device inside the flask). The bulb filtering device consists of a 5-mL pipet cut off at both ends: one end near the bulb and the bulb filled with glass wood and extending to within 1/4 inch of the bottom of the flask; the other end is left sufficiently long to extend 1-1/2 inches through the stopper and fitted with a short piece of rubber tubing (approximately 1 to 2 inches long).

(c) Procedure. Place approximately 30 mL of acetone saturated with an excess of B-1 dye specimen in the Erlenmeyer flask. Stir the saturated solution for 3-1/2 hours. Take a 1-mL aliquot by means of a 1-mL pipet fitted in the open end of the rubber tube and pass nitrogen through the right angle glass tube to force the saturated acetone solution up through the bulb filtering

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device into the 1-mL pipet. Dilute the 1-mL aliquot to 1000 mL with acetone in a volumetric flask at 30°C and mix well. Pipet a 5-mL aliquot into a 50-mL volumetric flask and make up to volume with acetone at 30°C. Using the same conditions as above for the preparation of the calibration curve, determine percent absorbance. After an additional 2-1/2 hours of stirring, a second 1-mL aliquot is taken from the saturated acetone solution and processed as above. If this absorbance reading is within 1.5 percent absorbance of the first aliquot, no further readings are made; otherwise, subsequent 1-mL aliquots are taken at 2-hour intervals and processed as above until 2 successive readings agree within 1.5 percent absorbance. No more than 3 aliquots shall be taken. From the calibration curve, determine the concentration and multiply by 10^4 to obtain concentration in grams per milliliter.

4.2.4.3 Melting point. Introduce a small portion of the B-1 dye into the open end of a melting point capillary tube (available as a commercial product or formed by sealing one end of a thin walled capillary tube approximately 90 millimeters (mm) long and 1.5 to 2.0 mm in diameter). Tap the sealed end of the melting point tube on a solid surface to pack the B-1 dye loosely in the sealed end of the tube to a height of 4 to 6 mm. Attach the tube to a calibrated mercury thermometer, graduated in intervals of no more than 0.5°C, in such a manner that the B-1 dye in the tube is adjacent to the midpoint of the thermometer bulb. Suspend the thermometer and melting point tube assembly in a circulating, clear, nonvolatile oil bath which has been preheated to 165°C. Adjust the heat so that the temperature increases at a rate of 1°C per minute. Record as the melting point that temperature at which the sample begins to melt. When necessary, apply a correction for the emergent stem of the thermometer.

4.2.4.4 Particle size.

(a) Retained on No. 325 sieve. Weigh 10.0 g of B-1 dye specimen to the nearest 0.01 g and transfer quantitatively to a 150-mL beaker containing approximately 75 mL of water and 1 mL of a 5-percent dispersing agent solution (see 6.4). Stir the mixture well and pour the mixture into a tared No. 325 sieve conforming to RR-S-366. Rinse the beaker and stirrer and pour the rinse water over the B-1 dye specimen in the sieve. Repeat the rinsing and washing until the water passing through the sieve is clear (contains no B-1 dye particles). Dry and weigh the sieve and determine the weight of the sample retained. Calculate particle size as follows:

$$\text{Percent retained on No. 325 sieve} = \frac{100B}{W}$$

where: B = Weight, in grams, retained on No. 325 sieve, and
W = Weight, in grams, of B-1 dye specimen.

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(b) Ten-micrometer and mean mass diameter. Determine the percent less than 10 micrometers and the mean mass diameter by means of a sedimentation procedure incorporating the principles of Stoke's Law (see 6.5).

4.2.4.5 Solubility in heptane. Weigh 1.00 g of B-1 dye specimen to the nearest 0.1 mg into a eight-ounce jar containing 120-mL of spectrophotometric grade heptane (Aldrich No. 15,487-3 or equal). Seal the jar with a screwcap and agitate for 30 minutes on a Red Devil Paint Shaker or similar device. Examine the solution for undissolved dye and repeat the test using a new batch of heptane if none is observed. If after the second test no residue remains, discontinue testing and consider the specimen as having failed the test. Filter solutions containing undissolved dye through a #1 Whatman (or equal) fluted filter paper. Pipet 1-, 2-, and 3-mL aliquots of the supernatant liquid into separate 25-mL volumetric flasks and label each. Make each up to the mark with spectrographic grade heptane. Measure the absorbance of each solution at 520 nanometers (nm) using a Beckman Model B spectrophotometer or equivalent and determine conformance to 3.1.1.1.

4.2.4.6 Infrared spectra. Weigh 1.0 mg of B-1 dye specimen to the nearest 0.1 mg and mix with 200 mg of potassium bromide (suitable for Infrared Spectroscopy) using a Spex Industries' "Wig-L-Bug" or "mixer/mill" or equal. Place the powder mix in a Perkins-Elmer Micro KBr Die or equal. Place the die in a Carver press or equal and press to 10,000 pounds per square inch for 2 minutes under vacuum. Remove the disc from the die and place in the holder for insertion into the sample window of a Perkins-Elmer Infrared Spectrophotometer No. 457 or equal. Scan the spectra at slow or medium speed with a normal slit opening. Major bands shall be as follows:

cm^{-1}	Strength
3420	Medium
1620	Strong
1510	Strong
1325	Very strong
1180	Strong
1135	Strong
1100	Strong
750	Strong

4.2.4.7 Purity. Weigh 0.1 g $\pm$ 0.001 g of the standard B-1 dye (6.6) and the B-1 dye test specimen into separate glass stoppered weighing bottles. With the stopper removed, dry specimens in an oven for 4 hours at 115° $\pm$ 5°C. Cool specimens in a desiccator and weigh. Wash the dyes into separate 1000 mL volumetric flasks and bring to total volume with spectrophotometric grade methanol. Determine the absorbance of each specimen with a Beckman DU Visible Spectrophotometer or equal at wavelengths of 512.5 nm and 368 nm. Calculate the molar absorptivity (E max) of the B-1 dye standard as follows:

$$E\ max = \frac{C\ B}{A}$$

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where: A = Absorbance,
B = Cell path length, and
C = Concentration.

Calculate the Absorbance Index (AI_{ss}) for the B-1 dye standard as follows:

$$AI_{ss} = \frac{Y}{Z}$$

where: Y = E max at 512.5 nm, and
Z = E max at 368 nm.

Calculate the molar absorptivity and absorbance index (AI_{ts}) for the test specimen using the equations above. Determine the purity of B-1 dye as follows:

$$\text{Purity of B-1 dye} = \frac{100AI_{ts}}{AI_{ss}}$$

5. PACKAGING

5.1 Unit packing, level C. The specified quantity (see 6.2) of dispersed formation of B-1 dye shall be unit packed in a steel drum having a nominal capacity not exceeding 15 gallons. The drum shall be furnished with a durable interior organic coating which shall neither affect nor be affected by the contents. The unit pack shall provide adequate protection against corrosion, deterioration, and damage from the supply source to the first receiving activity and for a minimum duration of one year. The unit pack shall comply with Department of Transportation (DOT) regulations, Uniform Freight Classification Rules, National Motor Freight Classification Rules, and any other carrier rules applicable to the mode of transportation. Containers shall be stackable and supportable of superimposed loads during shipment and shall assure safe delivery to destination without damage to contents.

5.2 Unitization. The unit packs of 5.1 shall be palletized in accordance with MIL-STD-147.

5.3 Marking. Unit packs and pallet loads shall be marked in accordance with Department of Transportation regulations and MIL-STD-129. Each container shall be marked to show the date of manufacture and lot number of the water dispersed formation of B-1 dye.

6. NOTES

6.1 Intended use. Water dispersed formation of B-1 dye covered by this specification is intended for use in the manufacture of B-1 paper, a component of the M9 Chemical Agent Detector Paper.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Quantity per pack required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Dispersants. Satisfactory dispersing agents may be obtained from the following companies:

Witko Chemical Company - 85 Flake
Monsanto Chemical Corporation - Maprofix-LK
Onyx Chemical Company - Onyxsan

6.5 Sedimentation balance. The following pieces of equipment have been used for determining particle size:

Sartorius Sedimentation Bal, Model No. 4600
Micromeritics Sedigraph 5000
Cahn Electrobalance, Model RG with No. 2800 particle sedimentation accessory

6.6 Standard B-1 dye. Standard B-1 dye is available from:

Commander
ARRADCOM
Chemical Systems Laboratory
ATTN: DRDAR-CLC-E
Aberdeen Proving Ground, MD 21010

6.7 Sampling and testing precautions. This specification covers inspection of chemical material (B-1 dye) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material.

6.8 Supersession data. This specification supersedes and includes the requirements of Edgewood Arsenal Purchase Description EA-D-1164, dated 13 June 1980.

Custodian:

Army - EA

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-51494(EA)	DYE, B-1, WATER DISPERSED FORMATION

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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